

Chapter

7

Fluid Dynamics

Fluid dynamics is the study of fluids in motion and their properties.

Two important conservation principles are used to simplify the analysis of fluid dynamics: conservation of mass and conservation of energy. The conservation of mass gives us the equation of continuity, while the conservation of energy is the basis of Bernoulli's equation.

VISCOSITY & FLUID VELOCITY:

"The resistance offered by different layers of a fluid to its flow called as viscosity".
Viscosity is a measure of how much a fluid resists flowing.

Fluids with low viscosity flow easily, while fluids with high viscosity flow slowly.

Coefficient of Viscosity: (η)

The coefficient of viscosity of a fluid is a measure of how thick and resistant to flow it is.

Fluids with low viscosity, like water, flow easily. Fluids with high viscosity, like honey, flow slowly.

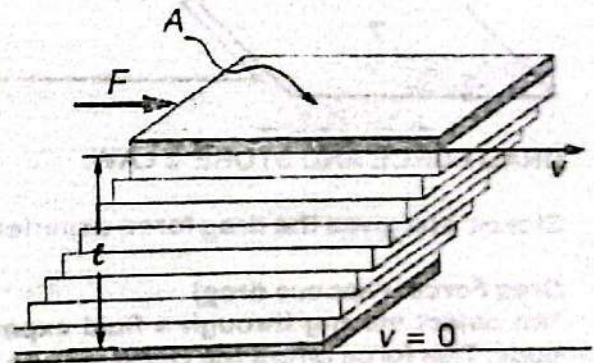
Examples of fluids with different viscosities:

- Water: low viscosity
- Syrup: medium viscosity
- Honey: high viscosity
- Engine oil: very high viscosity

Derivation:

In order to understand the coefficient of viscosity suppose we have two plane solid surfaces one above the other and between them is a viscous fluid.

The layer of the fluid in contact with the solid surface has nearly zero velocity. The next layer moves slowly over the first and the 2nd layers move with little more velocity on the first and so on. The two adjacent layers have a relative velocity (v) and opposing force (F) arises to stop their relative motion.



The opposing force (F) is directly proportional to velocity (v)

$$F \propto v \quad \text{---(1)}$$

The opposing force (F) is directly proportional to the area of cross-section (A)

$$F \propto A \quad \text{---(2)}$$

The opposing force (F) is inversely proportional to the length of layer

$$F \propto 1/L \quad \text{---(3)}$$

By combining all the equations we get:

$$F \propto \frac{Av}{l}$$

$$F = \eta \left(\frac{Av}{l} \right)$$

where η = Proportionality constant

η = is the coefficient of viscosity which depends upon the nature and temperature of the fluid.

(In liquid η decreases with increase in temperature)

In gases η increases with increase in temperature)

$$\text{For liquid } \eta \propto \frac{1}{T}, \text{ For gas } \eta \propto T$$

Above equation in terms of coefficient of viscosity is expressed as,

$$\eta = \frac{Fl}{AV}$$

i.e:

"The tangential force per unit area required to maintain a unit relative velocity between two layers which is unit distance apart".

S.I Unit:

The S.I Unit is (N-s)/m² or Pa-s.

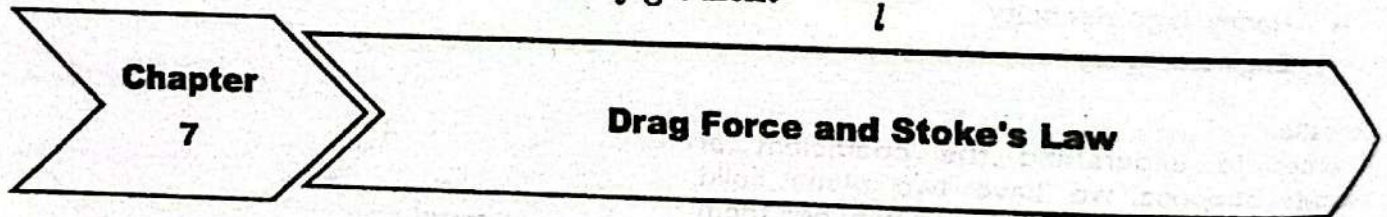
C.G.S unit : Poise = (dyne.sec)/cm².

Velocity Gradient ($\frac{v}{l}$):

In fluid dynamics velocity gradient is defined as:

"The rate of change of velocity with distance normal (perpendicular) to the direction of flow of the layers of the fluid with respect to the object passing through the fluid"

$$\text{Velocity gradient} = \frac{\Delta V}{l}$$



DRAG FORCE AND STOKE'S LAW:

Stokes' law gives the drag force experienced by a small spherical object through a fluid

Drag Force: (viscous drag)

"An object moving through a fluid experiences a retarding force just opposite to the weight of the body. This force offers the resistance to the motion of body".

The drag force increases as the speed of the object increases. The drag force depends upon the size, shape, relative speed of the object with respect to the fluid and viscosity of the fluid. The drag force on spherical object can be expressed by mathematical formula known as Stokes' law. Stokes did his research and study on spherical shaped object of uniform density. He formulated his work as follows:

$$F \propto \eta r v$$

So,

$$F = 6\pi \eta r v$$

where

F = Drag force (viscous drag)

η = Coefficient of viscosity

r = Radius of sphere.

v = velocity of the sphere.

TERMINAL VELOCITY:

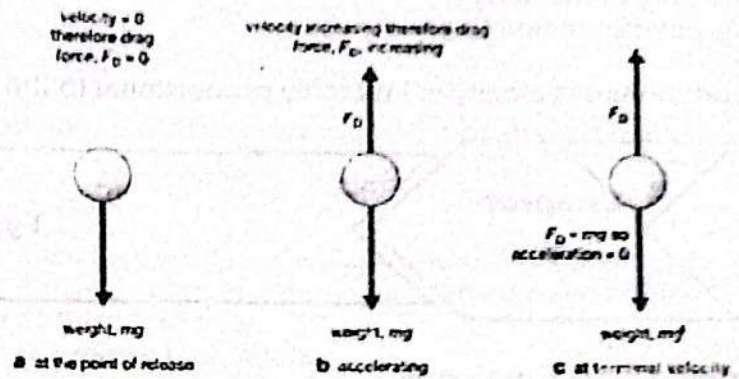
Terminal Velocity (V_t) refers to the constant velocity acquired by a freely falling body in a viscous medium.

Derivation:

To derive the formula for terminal velocity (V_t) of the spherical body consider a water droplet (assumed to be spherical) falling freely due to gravity in a viscous medium.

At point A the droplet is about to start its free fall. Here the net force will be equal to the weight of the droplet. When an object falls through a fluid it will experience a dragging force (F_d). as the object falls faster and faster the drag force increases as shown in the diagram.

Eventually the drag force increases to a value where it is equal to the weight of the object (spherical body) and it continues to fall at a steady speed (constant speed) we call this terminal velocity (V_t).



Now at the terminal velocity

$$F_d = mg$$

By Stoke's Law we have:

$$F_d = 6\pi \eta r v$$

Where v = terminal velocity = V_t

By comparing both equations we get:

$$6\pi \eta r V_t = mg$$

$$V_t = \frac{mg}{6\pi \eta r}$$

For a sphere of uniform density:

$$m = \rho V$$

ρ -- Density of droplets

V -- Volume of object

Also,

$$V = \frac{4}{3}\pi r^3$$

Above equation becomes,

$$m = \frac{4}{3}\pi r^3 \rho$$

Putting the value of m we get,

$$V_t = \frac{\frac{4}{3}\pi r^3 \rho g}{6\pi \eta r}$$

Equation shows the terminal velocity of spherical body moving through a viscous medium.

$$V_t = \frac{2}{9} \frac{(r^2 \rho g)}{\eta}$$

Conclusion:

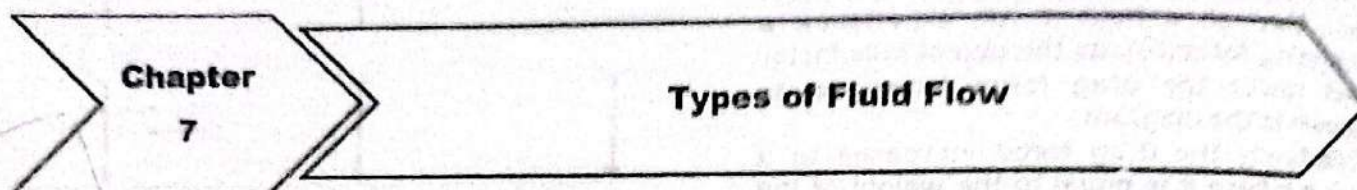
From equation, we can conclude that terminal velocity is directly proportional to

Square of the radius of the body (r^2)

Density of the body (ρ)

Acceleration due to gravity (g)

But terminal velocity is inversely proportional to the coefficient of viscosity (η)



TYPES OF FLUID FLOW:

When a fluid is in motion its flow can be characterized in various ways which are as follow.

- Steady and Unsteady Flow
- Laminar and Turbulent Flow
- Compressible and Incompressible flow

Steady and Unsteady Flow:

Steady Flow: A steady flow is defined as when its fluid characteristics like velocity, density and pressure at a point, do not change with time. A steady flow can be mathematically expressed as,

$$\frac{\Delta P}{\Delta t} = 0, \frac{\Delta V}{\Delta t} = 0, \frac{\Delta \rho}{\Delta t} = 0$$

Example:

Water flowing through a tap

Unsteady Flow: A flow is defines as unsteady when the fluid characteristics like velocity, density and pressure at a point changes with respect to time. This can be mathematically explained as,

$$\frac{\Delta P}{\Delta t} \neq 0, \frac{\Delta V}{\Delta t} \neq 0, \frac{\Delta \rho}{\Delta t} \neq 0$$

Example:

Water waves, Pulsatile, blood flow, pipe filling and draining etc.

Compressible and Incompressible Flows:

Compressible Flows: A compressible flow is that type of flow in which the density of the fluid changes from one point to another point. This means that density is not constant throughout the flow.

Example: gas, vapor and steam etc.

Incompressible Flow:

Incompressible flow is the type of flow in which the density of the liquid is constant throughout the flow.

An incompressible, non-viscous fluid is called an ideal fluid. Although no real fluid have zero viscosity at room temperature but some fluids have very low viscosity values (negligible).

Example: Nearly all liquids are incompressible.

Laminar (Stream Line) and Turbulent Flow:

Laminar Flow: Laminar or streamline flow is defined as a type of flow in which the liquid particles move along a well-defined streamline or paths, such that all the stream lines are straight and parallel to each other. Different stream lines or laminar cannot cross each other under this flow condition. The layers in laminar flow glide smoothly over the adjacent layer. The flow is laminar when the Reynolds Number is less than 2300.

The key features of laminar flow is:

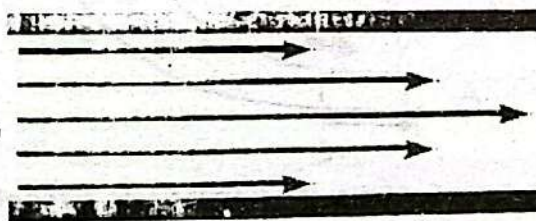
- Streamline flow
- Smooth velocity profile
- Low mixing and diffusion
- Low shear stress
- Laminar flow is commonly observed in situations with low rates, small pipe diameter and high fluid viscosity.

Turbulent Flow:

The flow of liquid becomes irregular or turbulent above a certain velocity or under any conditions that can cause abrupt changes in velocities. The particles of fluid move in zig-zag manner results in high eddies formed, such type of flow is referred as turbulent flow. The Flow is turbulent when the Reynolds number is greater than 4000.

The key features of turbulent flow are:

- High Reynolds Number
- Rough surfaces occur turbulent flow
- High Velocity
- Pressure gradient
- Agitation and stirring in Industries



Laminar Flow



Turbulent Flow

Transition from laminar to turbulent Flow:

The flow of fluid deviates from its streamlined path when the velocity of flow increases to cause turbulence. The state of flow between laminar and turbulence is considered to be a transition flow, which can commence at a random point during the flow.

The main parameter characterizing transition is the Reynolds Number.

Reynolds Number:

In Fluid mechanics, the Reynolds number (Re) is the dimensionless quantity that helps predict fluid flow patterns (Laminar/Turbulent) in different situations by measuring the ratio between inertial and viscous forces.

Significance of Reynolds Number:

The Reynolds Number has great importance to give the criterion of the fluid flow is streamlined or turbulent.

- At low Reynolds number ($Re < 2300$):
Viscous effect is comparable to inertial effect and flow behaves in orderly manner (laminar flow)
- At high Reynolds number ($Re > 4000$):

Viscous effect is insignificant compared to inertial effect in high Reynolds number fluid

- Flow pattern is irregular, unsteady and random (turbulent flow)
- R_e between 2300 and 4000:
Transient flow where the flow velocity and pressure changes with time.

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Equation of Continuity

EQUATION OF CONTINUITY:

Equation of continuity is the mathematical expression of the principle of conservation of mass

Statement:

It states that: For steady state flow the mass of the flow into the volume must be equal to the mass flow rate out.

Derivation.

Consider a fluid having mass m flowing through a pipe of non-uniform cross sectional areas. The fluid of same mass m is flowing steadily between two stationary cross sectional areas A_1 and A_2 , where $A_1 > A_2$.

Distance travelled by fluid in a very small interval of time Δt when it flows through A_1 is

$$\Delta l_1 = v_1 \Delta t$$

The Volume of fluid flow through A_1 at time Δt is

$$\Delta V_1 = \Delta l_1 \times A_1$$

$$\Delta V_1 = A_1 v_1 \Delta t$$

Similarly, Distance travelled by fluid in a very small interval of time Δt when it flows through A_2 is

$$\Delta l_2 = v_2 \Delta t$$

The Volume of fluid flow through A_2 at time Δt is

$$\Delta V_2 = \Delta l_2 \times A_2$$

Put the values of Δl_1 and Δl_2 in above equations of volume we get:

$$\Delta V_2 = A_2 v_2 \Delta t$$

Now the mass of incompressible fluid of density ρ flowing through A_1 in time Δt is,

$$\Delta m_1 = \rho \Delta V_1$$

As we know

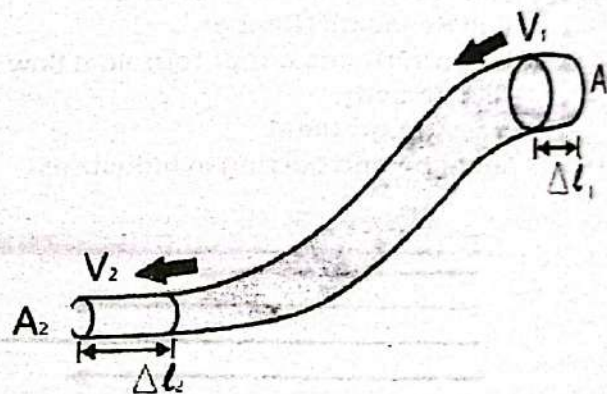
$$\Delta V_1 = A_1 v_1 \Delta t$$

then

$$\Delta m_1 = \rho A_1 v_1 \Delta t \quad \text{----- (i)}$$

Similarly, the mass of incompressible fluid of density ρ flowing through A_2 in time Δt is,

$$\Delta m_2 = \rho \Delta V_2$$



$$\therefore [\Delta V_2 = A_2 v_2 \Delta t]$$

$$\therefore \Delta m_2 = \rho A_2 v_2 \Delta t \quad \text{---(ii)}$$

In steady flow the total mass of the tube remains the same so,
 $\Delta m_1 = \Delta m_2$

$$\rho A_1 v_1 \Delta t = \rho A_2 v_2 \Delta t$$

$$\therefore A_1 v_1 = A_2 v_2$$

This is the equation of continuity for the steady state flow of incompressible fluid.
 Thus we know,

$$(\text{Area of pipe}) \times (\text{velocity of fluid}) = \text{constant}$$

This constant is equal to time rate of volume of liquid passed through the pipe thus,

$$Av = \frac{\Delta V}{\Delta t} = \text{constant}$$

Now combining the above two equations (i) and (ii) we can say that :

$$\frac{\Delta m}{\Delta t} = \rho \frac{\Delta V}{\Delta t}$$

Conclusion:
 From Equation:

$$\frac{A_1}{A_2} = \frac{v_2}{v_1}$$

The speed of fluid through any pipe is inversely proportional to the cross sectional area of the pipe.

$$v \propto \frac{1}{A}$$

Physical Significance:

The deep part of a river has larger cross sectional area in comparison to the shallow part of the rivers, thus deep water flows slowly in comparison to shallow water.

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Bernoulli's Equation

BERNOULLI'S EQUATION:

Introduction:

As a fluid flows through a pipe of varying cross-sectional Area and elevation the pressure changes along the pipe. In 1738 a Swiss Physicist Daniel Bernoulli derived an expression that relates the pressure of a fluid to its speed and elevation. This equation or theorem is a consequence of law of conservation of energy.

Assumptions.

For the derivation of Bernoulli's equation we assume that

- The fluid flow must be steady and streamline
- The fluid is incompressible
- Density of the fluid must be constant throughout the flow
- There are no viscous forces nor friction.

Physics XI

The relationship between the pressure, flow speed and height for flow of an ideal fluid is called Bernoulli's Equation.

Derivation:

Consider an element of fluid that lies between two cross sectional Areas A_1 and A_2 as shown in the diagram

Let V_1 be the speed of incompressible fluid at A_1
 a. The distance Δx_1 travelled by the fluid in a very short time Δt while passing through area A_1 is,

$$\Delta x_1 = V_1 \Delta t$$

Similarly, V_2 be the speed of A_2 and the distance Δx_2 travelled by the fluid in short interval of time it is,

$$\Delta x_2 = V_2 \Delta t$$

As the fluid is incompressible. Thus, according to equation of continuity,

$$\frac{\Delta V}{\Delta t} = A v$$

As we know,

$$\therefore v = \frac{\Delta x}{\Delta t}$$

Therefore,

$$\frac{\Delta V}{\Delta t} = A \frac{\Delta x}{\Delta t}$$

For Area A_1 and A_2 above equation becomes,

$$\frac{\Delta V}{\Delta t} = \frac{A_1 \Delta x_1}{\Delta t} = \frac{A_2 \Delta x_2}{\Delta t}$$

Or

$$\Delta V = A_1 \Delta x_1 = A_2 \Delta x_2$$

Where

ΔV is the volume of fluid passing through cross-sectional Area A_1 and A_2 in time interval Δt .
 To calculate the work done on a fluid element during time interval Δt .

Now Suppose,

P_1 is the pressure at A_1 ,

$F_1 = P_1 A_1$ is the force on A_1

P_2 is the pressure at A_2 ,

$F_2 = P_2 A_2$ is the force on A_2

The Work done at A_1 is

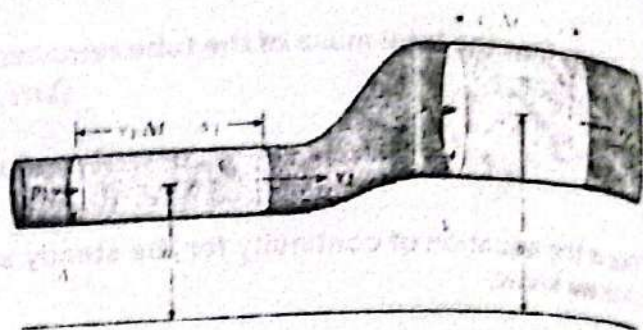
$$\Delta W_1 = F_1 \Delta x_1$$

Or

$$\Delta W_1 = P_1 A_1 \Delta x_1$$

And the work done by the fluid at A_2 is,

$$\Delta W_2 = -F_2 \Delta x_2 = -P_2 A_2 \Delta x_2$$



This work ΔW_2 is negative because the force F_2 at A_2 is opposite to the direction of displacement.

Now the net work done of fluid from A_1 to A_2 is,

$$\Delta W = \Delta W_1 + \Delta W_2$$

Putting values of ΔW_1 and ΔW_2 we get;

$$\begin{aligned}\Delta W &= P_1 A_1 \Delta x_1 + (-P_2 A_2 \Delta x_2) \\ &= P_1 A_1 \Delta x_1 - P_2 A_2 \Delta x_2\end{aligned}$$

As we know,

$$A_1 \Delta x_1 = A_2 \Delta x_2 = A \Delta x$$

Substituting we get,

$$\begin{aligned}\Delta W &= P_1 A \Delta x - P_2 A \Delta x \\ \Delta W &= (P_1 - P_2) A \Delta x,\end{aligned}$$

As

$$A \Delta x = \Delta V$$

Therefore,

$$\Delta W = (P_1 - P_2) \Delta V$$

If m is the mass of the fluid having volume ΔV then ρ density of the fluid will be,

$$\begin{aligned}\rho &= \frac{m}{\Delta V} \\ \Rightarrow \Delta V &= \frac{m}{\rho}\end{aligned}$$

Above Equation becomes,

$$\Delta W = (P_1 - P_2) \frac{m}{\rho}$$

The fluid is falling from height h_1 from cross sectional area A_1 to cross-sectional Area A_2 at height h_2 from the surface of the Earth. Thus, the part of work done is used in changing K.E of fluid and a part of it is used in changing gravitational potential energy of fluid. So,

$$\Delta W = \Delta(K.E) + \Delta(P.E)$$

So,

$$(P_1 - P_2) \frac{m}{\rho} = \left[\frac{1}{2} m v_2^2 - \frac{1}{2} m v_1^2 \right] + [(mgh_2 - mgh_1)]$$

$$(P_1 - P_2) \frac{m}{\rho} = \frac{1}{2} m (v_2^2 - v_1^2) + m (gh_2 - gh_1)$$

$$(P_1 - P_2) \frac{1}{\rho} = \frac{1}{2} (v_2^2 - v_1^2) + (gh_2 - gh_1)$$

$$P_1 - P_2 = -\frac{1}{2} \rho v_1^2 + \frac{1}{2} \rho v_2^2 + \rho gh_2 - \rho gh_1$$

$$P_1 + \frac{1}{2} \rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2} \rho v_2^2 + \rho gh_2$$

This is known as Bernoulli's Equation and its often expressed generally as,

$$P + \frac{1}{2} \rho v^2 + \rho gh = \text{Constant}$$

Statement:

Bernoulli's equation/theorem states that the

"Sum of pressure, kinetic energy per unit volume and potential energy per unit volume has the same value at all points along a streamline motion".

$$P + \frac{1}{2} \rho v^2 + \rho gh = \text{Constant}$$

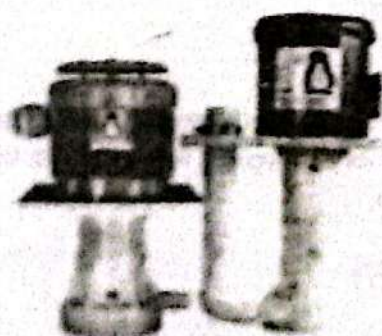
Where,

$\frac{1}{2} \rho v^2$ is dynamic pressure

ρgh is static pressure

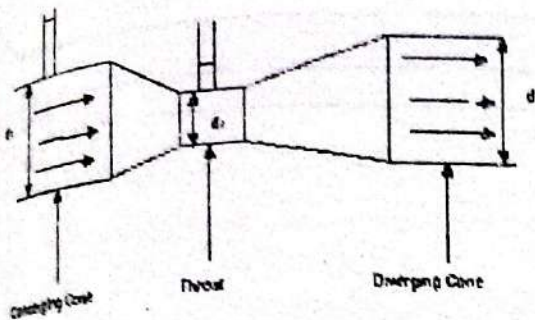
APPLICATION OF BERNOULLI PRINCIPAL

Filter pump



A filter pump is a device that uses Bernoulli's principle to create a vacuum to draw liquid through a filter. The pump works by creating a high-speed jet of air, which creates a low-pressure region behind it. This low-pressure region draws liquid through the filter.

Venturi meter



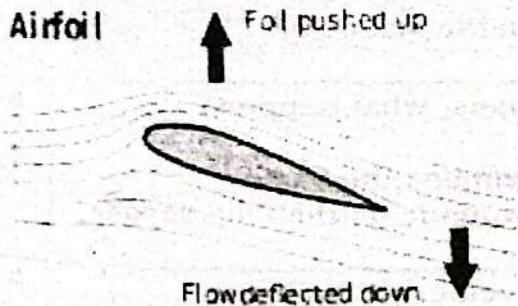
A Venturi meter is a device that uses Bernoulli's principle to measure the flow rate of fluids. The meter consists of a tube with a narrowed section in the middle. As the fluid flows through the narrowed section, its speed increases and its pressure decreases. This pressure difference is measured by two pressure taps, one upstream and one downstream of the narrowed section. The flow rate of the fluid can then be calculated from the pressure difference.

Atomizer



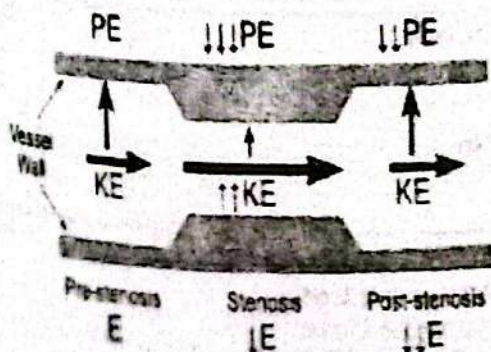
An atomizer is a device that uses Bernoulli's principle to create a fine mist of liquid. The liquid is pumped through a nozzle at high speed, which creates a low-pressure region behind it. This low-pressure region draws air into the atomizer, which mixes with the liquid to create a mist.

Flow of air over an aero foil



The wings of an airplane are shaped so that the air flows faster over the top of the wing than over the bottom of the wing. This creates a pressure difference between the top and bottom of the wing, which lifts the airplane up. This is known as lift.

Blood Physics



Bernoulli's principle also plays a role in blood flow. The heart pumps blood through the arteries at high speed. This creates a low-pressure region in the veins, which draws blood back to the heart.

Chapter

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Multiple Choice Questions

MCQ'S		ANS.
1.	For an incompressible fluid, the flow rate is (a) equal for all surfaces. (b) constant throughout the pipe. (c) greater for the larger parts of the pipe. (d) none of the above	b
2.	Bernoulli's principle states that for horizontal flow of a fluid through a tube, the sum of the pressure and energy of motion per unit volume is (a) increasing with time (c) constant (c) decreasing with time (d) varying with time	b
3.	Which of the following is associated with the law of conservation of energy in fluids? (a) Archimedes' principle (b) Pascal's principle (c) Bernoulli's principle (d) equation of continuity	c
4.	As the speed of a moving fluid increases, the pressure in the fluid (a) increases (b) remains constant (c) decreases (d) may increase or decrease, depending on the viscosity	c
5.	If the cross-sectional area of a pipe decreases, what happens to the fluid velocity? (a) Increases (b) Remains the same (b) Decreases (d) Depends on the fluid density	a
6.	A sky diver falls through the air at terminal velocity. The force of air resistance on him is (a) half his weight (b) equal to his weight (c) twice his weight (d) Cannot be determined from the information given.	b
7.	Wind speeding up as it blows over the top of a hill (a) Increases atmospheric pressure there. (b) decreases atmospheric pressure there. (c) doesn't affect atmospheric pressure there. (d) equal's atmospheric pressure.	b
8.	A fluid is undergoing "Incompressible" flow. This means that: (a) the pressure at a given point cannot change with time (b) the velocity at a given point cannot change with time (c) the velocity must be the same everywhere (d) the pressure must be the same everywhere (e) the density cannot change with time or location	a

9.	A fluid is undergoing steady flow. Therefore: (a) the velocity of any given molecule of fluid does not change (b) the pressure does not vary from point to point (c) the velocity at any given point does not vary with time (d) the density does not vary from point to point	c
10.	The equation of continuity for fluid flow can be derived from the conservation of: (a) energy (b) volume (c) mass (d) pressure	c

Exams Practice MCQs

1.	Which of the following is NOT a type of fluid dynamics? (a) Aerodynamics (b) Hydrodynamics (c) Magnetohydrodynamics (d) Electrodynamics	d
2.	Which of the following is a type of fluid flow? (a) Laminar flow (b) Turbulent flow (c) Steady flow (d) All of the above	d
3.	Which of the following is NOT a factor that affects the viscosity of a fluid? (a) Temperature (b) Pressure (c) Density (d) Gravity	d
4.	The coefficient of viscosity is a measure of: (a) The resistance of a fluid to flow (b) The force required to move a fluid (c) The velocity of a fluid (d) The pressure of a fluid	a
5.	The SI unit of the coefficient of viscosity is: (a) Pascal seconds (Pa.s) (b) Newton second per meter square (N.s/m ²) (c) Poise (PI) (d) All of the above	d
6.	Which of the following fluids has the highest coefficient of viscosity? (a) Water (b) Honey (c) Air (d) Alcohol	b
7.	The coefficient of viscosity of a fluid decreases with: (a) Temperature (b) Pressure (c) Density (d) Molecular structure	a
8.	Which of the following methods can be used to measure the coefficient of viscosity of a fluid? (a) Capillary viscometer (b) Falling ball viscometer (c) Rotational viscometer (d) All of the above	d
9.	The terminal velocity of a spherical body falling through a fluid is inversely proportional to: (a) The radius of the body (b) The density of the body (c) The viscosity of the fluid (d) All of the above	c

10. Which of the following statements about the Reynolds number is CORRECT? (a) The Reynolds number is directly proportional to the fluid velocity and inversely proportional to the fluid viscosity. (b) The Reynolds number is directly proportional to the fluid viscosity and inversely proportional to the fluid velocity. (c) The Reynolds number is independent of both the fluid velocity and viscosity. (d) None of the above	a
11. Which of the following flows is MOST likely to be laminar? (a) Flow of water through a pipe with a large diameter at a high velocity (b) Flow of honey through a pipe with a small diameter at a low velocity (c) Flow of air through a jet engine at a high velocity (d) None of the above	b
12. Which of the following flows is MOST likely to be turbulent? (a) Flow of water through a pipe with a large diameter at a low velocity (b) Flow of honey through a pipe with a small diameter at a high velocity (c) Flow of air through a jet engine at a high velocity (d) None of the above	c
13. If the density is 1000 kg/m^3 , the velocity is 2 m/s , the pipe diameter is 0.1 m , and the dynamic viscosity is $0.001 \text{ Pa}\cdot\text{s}$, what is the Reynolds number? (a) 400 (b) 2000 (c) 20,000 (d) 4,000,000	b
14. In an experiment, a fluid with a dynamic viscosity of $0.02 \text{ Pa}\cdot\text{s}$ flows through a pipe with a diameter of 0.05 m at a velocity of 2 m/s . If the density of the fluid is 800 kg/m^3 , what is the Reynolds number for this flow? (a) 40 (b) 80 (c) 800 (d) 4000	d
15. In fluid dynamics, a Reynolds number less than approximately 2000 is often indicative of which type of flow? (a) Laminar flow (b) Turbulent flow (c) Transitional flow (d) Unsteady flow	a
16. Which of the following changes will increase the Reynolds number? (a) Increasing the density of the fluid (b) Increasing the viscosity of the fluid (c) Increasing the diameter of the pipe (d) Increasing the velocity of the fluid	b
17. Which of the following factors contribute to the prevalence of turbulent flow in practical scenarios? (a) High Reynolds numbers (b) Rough surfaces (c) High velocities (d) All of the above	d
18. Which of the following is an example of a real-world application of turbulent flow? (a) Flow of blood through arteries and veins (b) Flow of air over the wings of an airplane (c) Flow of water through a fire hose (d) All of the above	d
19. A pipe with a cross-sectional area of 100 cm^2 carries water at a velocity of 20 cm/s . What is the mass flow rate through the pipe, if the density of water is 1000 kg/m^3 ? (a) 2 kg/s (b) 20 kg/s (c) 200 kg/s (d) 2000 kg/s	a

20. A fluid is flowing through a pipe with a cross-sectional area of 50 cm^2 at a velocity of 10 cm/s . If the cross-sectional area of the pipe decreases to 25 cm^2 , what is the new velocity of the fluid, assuming that the mass flow rate remains constant? (A) 5 cm/s (B) 10 cm/s (C) 20 cm/s (D) 40 cm/s	c
21. What is the consequence of the equation of continuity? (a) The mass flow rate through any section of a pipe must be constant. (b) The velocity of a fluid must increase as the cross-sectional area of the pipe decreases. (c) The density of a fluid must decrease as the velocity of the fluid increases. (d) The density of a fluid must increase as the velocity of the fluid increases.	a
22. What is the consequence of the Bernoulli equation? (A) The sum of the pressure, velocity, and height terms in the Bernoulli equation must be constant along a streamline. (B) The pressure of a fluid must decrease as the velocity of the fluid increases. (C) The height of a fluid must increase as the pressure of the fluid decreases. (D) All of the above	a
23. A fluid is flowing through a pipe at a velocity of 10 m/s . If the pipe narrows to a section where the velocity increases to 20 m/s , what is the pressure change between the two sections, assuming that the fluid is incompressible? (a) 150 Pa (b) 15000 Pa (c) $150,000 \text{ Pa}$ (d) 1500 Pa	c
24. Which of the following is an example of an application of Bernoulli's principle? (a) Venturi meter (b) Airplane wing (c) Filter pump (d) All of the above	d
25. The venturi effect is a phenomenon in which the fluid pressure in a pipe decrease when the fluid velocity increases. Which of the following is NOT a consequence of the venturi effect? (a) The fluid velocity increases in the narrower section of a venturi meter. (b) The fluid pressure decreases in the wider section of a venturi meter. (c) The fluid density increases in the wider section of a venturi meter. (d) The fluid velocity decreases in the narrower section of a venturi meter.	c
26. What is the purpose of an atomizer? (a) To disperse a liquid into a fine spray. (b) To filter a liquid to remove impurities. (c) To measure the flow rate of a liquid. (d) To increase the pressure of a liquid	a
27. What is the primary purpose of the curved surface known as an airfoil? (a) To reduce air resistance (b) To increase the weight of the aircraft (c) To generate lift when moving through the air (d) To improve fuel efficiency	c
28. In the Bernoulli equation, the term ρgh represents: (a) Total pressure (b) Static pressure (c) Dynamic pressure (d) Fluid pressure	b
29. Which of the following is NOT an assumption of the Bernoulli equation? (a) The flow must be steady and streamline.	b

- (b). The fluid is compressible.
- (c). There are no viscous forces in the fluid, and friction is negligible.
- (d). The density of the fluid remains constant at all points during the flow.

30. Which of the following is NOT a key characteristic of laminar flow?

- (A). Streamline flow
- (B). Smooth velocity profile
- (C). High mixing and diffusion
- (D). Predictable flow behavior

c

31. Stokes' law states that the drag force on a small spherical object moving through a viscous fluid is proportional to which of the following?

- (a). Velocity of the object
- (b). Radius of the object
- (c). Viscosity of the fluid
- (d). All of the above

d

Viscous Force in a Fluid	$F = \eta A \cdot \frac{V}{l}$
Stoke's Law	$F_{\text{drag}} = 6\pi\eta r v$
Terminal Velocity of Sphere in a Fluid	$v = \frac{2}{9} \frac{r^2 \rho g}{\eta}$
Kinematics Viscosity	$\text{Kinematics Viscosity} = \frac{\text{Dynamic Viscosity}}{\text{fluid density}}$
Reynold's Number	$\text{Rynold's no.} = \frac{\text{fluid velocity} \times \text{Internal Diameter}}{\text{Kinematics Viscosity}}$ $R_e = \frac{\rho \bar{V} L}{\eta}$
Velocity of liquid flow	$v = \frac{R_e \eta}{2\rho}$

Physics XI

Equation of continuity

$$\frac{\Delta V}{\Delta t} = Av = \text{constant}$$

Bernoulli's Equation

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

of equal size are falling through air at a vel